

Modern JavaScript Operations

Explore core JavaScript concepts through interactive operations without DOM manipulation



Array Operations

Perform various array manipulations including:

- Adding/removing elements (push, pop, shift, unshift)
- Slicing and splicing arrays
- Transforming with map, filter, reduce
- Combining and copying arrays
- Searching and sorting elements

Run Array Operations



String Operations

Manipulate and analyze text strings including:

- Changing case (upper/lower)
- Searching within strings
- Extracting substrings
- Splitting and joining
- Replacing text patterns
- String formatting

Run String Operations



Object Operations

Work with JavaScript objects including:

- Creating and modifying objects
- Accessing nested properties
- Object iteration methods
- Getting keys/values
- Object copying and merging
- JSON serialization

Run Object Operations



Function Operations

Explore JavaScript functions including:

- Function declarations vs expressions
- Arrow functions
- Higher-order functions
- Closures and private variables
- Callback patterns
- Recursive functions

Run Function Operations



Data Processing

Process sample datasets including:

- Product inventory analysis
- Employee records processing
- Data aggregation
- Statistical calculations
- Grouping and categorization
- Data transformation

Run Data Processing



Search Operations

Search through data including:

- Text search in strings
- Finding array elements
- Object property search
- Case-insensitive search
- Partial match search
- Multi-criteria search

Run Search Operations



Sort Operations

Sort data in various ways:

- Alphabetical sorting
- Numerical sorting
- Custom sort functions
- Ascending/descending order
- Multi-property sorting
- Localized string sorting

Run Sort Operations



Filter Operations

Filter datasets based on criteria:

- Range filtering (numbers, dates)
- Category filtering
- Multiple condition filters
- Text content filters
- Custom filter functions
- Chaining filters

Run Filter Operations



Math Operations

Perform mathematical calculations:

- Basic arithmetic
- Advanced math functions
- Random number generation
- Rounding operations
- Exponents and roots
- Trigonometric functions

Run Math Operations



Date Operations

Work with dates and times:

- Date creation and parsing
- Date component access
- Date formatting
- Date arithmetic
- Time differences
- Date comparisons

Run Date Operations



Advanced Functions

Explore advanced function concepts:

- Recursive functions
- Memoization
- Currying
- Function composition
- Promises (simulated)
- Higher-order patterns

Run Advanced Functions



Exit Application

Close the application:

- Confirm before exiting
- Display farewell message
- Clean up resources

Exit Application

>_ Operation Output

This application now uses `alert()` and `prompt()` for all interactions instead of DOM manipulation.